

Ronald Sin . Jarvis Consulting

I am a software developer with a B.Sc. in Computer Science (Co-op) from Ontario Tech University. At Jarvis, I design and implement various backend projects and data pipelines using tools such as Python and Java. One recent project involved analyzing SQL retail data with NumPy and Pandas to help a gift shop maximize its revenue. Previously, I taught TypeScript and coding concepts such as variables and data structures at Code Ninjas, earning consistently high reviews from children and parents. Recently, I have been developing my data engineering skills after a hackathon sparked my interest in the field. I love data engineering because it combines my passion for problem-solving with the puzzle of transforming messy, unstructured data into clean, actionable insights.

Skills

Proficient: Python (NumPy, SciPy, Pandas), Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git

Competent: JUnit, JavaScript, C#, Databricks, Maven

Familiar: Flutter, Unity, HTML, CSS, React

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_RonaldSin

Cluster Monitor [GitHub]: Developed a metrics collection and storage system for Linux cluster monitoring. Leveraged Bash scripts to extract, process, and store system and hardware performance metrics. Deployed a Dockerized PostgreSQL database for persistent data storage. Streamlined deployment workflows through automated database provisioning and management scripts, reducing setup time. Scheduled automated data collection using Crontab.

Java Grep App [GitHub]: Developed a Java application modeled after the Linux grep utility that recursively searches target directories to output lines matching a user-specified regex pattern to a destination file. Utilized Java lambdas and streams to scale the application and efficiently handle large file sizes. Built a robust test suite using JUnit, managed project dependencies with Maven, and containerized the application with Docker to publish it directly to Docker Hub for seamless deployment.

Python Analytics Project [GitHub]: Ingested two years of anonymous retail data from a UK-based gift shop into a Dockerized PostgreSQL database, then cleaned and transformed the dataset using Python, NumPy, and Pandas to identify strategies for driving stagnant revenue. Based on these findings, designed and delivered a stakeholder presentation that highlighted a major, untapped non-home market and outlined actionable next steps to capitalize on the opportunity.

Highlighted Projects

Image Filter and Effects App [GitHub]: Developed a Python-based photo editing tool with targeted colour isolation and replacement capabilities. Manipulated colour space data using NumPy and SciPy matrix transformations to isolate and replace specific colour ranges within images. Integrated a fuzzy selection feature with adjustable sensitivity thresholds for precise colour targeting. Implemented lossless upscaling algorithms based on mathematical interpolation techniques to preserve image fidelity. Designed an intuitive PySimpleGUI interface to enhance accessibility for non-technical users.

Hospital Staffing Level AI Prediction Tool [GitHub]: Led a team of three developers in a high-pressure hackathon environment, coordinating task delegation and sprint planning to deliver a Granite AI-powered staffing optimization tool for healthcare settings. Engineered a robust Python data pipeline to validate and standardize input data, ensuring consistency across the system. Implemented advanced prompt engineering techniques to fine-tune Granite AI responses, achieving a 25%+ reduction in average simulated patient wait times.

Professional Experiences

Software Developer, Jarvis (2026-present): Designed and delivered a diverse set of data engineering projects spanning a wide range of topics. Actively participated in Scrum ceremonies including daily stand-ups, sprint retrospectives, and sprint reviews to ensure iterative progress. Consistently incorporated Scrum Master feedback to refine technical skills and elevate code quality standards. Worked with colleagues through code reviews and study sessions, improving overall code quality and conceptual understanding.

Tester and Computer Science Instructor, Code Ninjas (2022): Developed over 100 test cases and discovered and assisted in fixing over 50 bugs for an in-house booking web tool. Maintained over 25 computer workstations, contributing

to nearly 100% uptime of facility hardware and software. Managed a team of 23 employees performing coordination, task delegation, and administrative duties. Taught TypeScript and coding concepts such as variables, functions, and data structures to children, consistently receiving positive reviews and maintaining high client retention rates.

Model Maker and Industrial Painter, PCL Graphics (2025): Fabricated luxurious, 3D interactive display pieces such as an LED-illuminated selfie booth for high-profile clients in the cosmetics industry, maintaining strict attention to detail to ensure near-perfect finishes. Worked with a wide variety of materials, including wood, metal, and 3D prints, and applied industrial spray paint to complex shapes to achieve precise colour matches and consistent finishes. Researched, developed, and presented prototypes and proofs of concept to project leads, clearly communicating the pros, cons, technical risks, and proposed solutions for each design. Collaborated closely with client-facing teams to establish flexible project plans that balanced budgets, deadlines, and changing requirements, while quickly troubleshooting and resolving critical design flaws. Additionally, coordinated across multiple departments to streamline the procurement and processing of raw materials.

Education

University of Ontario Institute of Technology (Ontario Tech University) (2019-2024), Bachelor of Science, Computer Science Co-op

Miscellaneous

- **Band Member** - I participate in weekly band practices as a guitarist, jamming and creating covers of various rock and J-Rock songs. I'm constantly scouring Facebook Marketplace for deals on new gear and always on the lookout for fresh music. Some of my favourite bands include Broken By The Scream, Midori, and Imperial Circus Dead Decadence.
- **Motorcycles** - I look forward to nice weather on my days off so I can take my Yamaha R3 out for a ride. The full-body engagement and lack of physical barriers allow me to experience the smells and sights of country roads in ways a car never could.
- **Volleyball Player** - I'm a casual recreational player who enjoys pickup volleyball matches with friends. I love the fast-paced, team-oriented gameplay where every player must execute their role for the team to succeed. The post-game meals with everyone are always a highlight.